MATH118 MWF 9:00–9:50 AM, Room: Public Affairs Building, Room 1222

Instructor: Daniel Mckenzie, Math. Sciences (MS) 5346, mckenzie@math.ucla.edu

Teaching Assistant: Yuejiao Sun, MS 7630, sunyj@math.ucla.edu

Office Hours:

- Daniel Mckenzie: Monday 1–2pm, Thursday 1–2pm or by appointment.
- Yuejiao Sun: Tuesdays 10am–12.

Textbooks:

1. Elden, Lars. *Matrix Methods in Data Mining and Pattern Recognition.* The Society for Industrial and Applied Mathematics, 2019.

2. Chong, E and Zak, S. *An Introduction to Optimization, 4th Edition.* Wiley, 2013.

3. Hillier, Frederick S. and Lieberman, Gerald J. *Introduction to Operations Research, 10th Edition.* McGraw-Hill Higher Education, 2014.

We will be using the first two books quite a lot, so I recommend you find a way to access them. We will be using the third book sparingly, so I don't think you will need to buy a copy of it.

Prerequisite: Math 115A and some experience with scientific computing in MATLAB or Python. Note that on CCLE Math 42 is listed as a prerequisite, but this is not essential.

Course Objectives: This course will cover the key techniques from linear algebra and optimization that underlie much of modern data science. While there will be an emphasis on theory, we will always develop it with an eye towards applications. Theoretical analysis of algorithms will be complemented with programming exercises, in Python or MATLAB, that implement these algorithms to solve problems of interest.

Communication: This class will use Piazza for classroom discussion. If you have questions about the course content (*e.g.* "Where do I start with question 14 on the homework?" or "Can someone explain that thing we did at the end of class yesterday?") or of an administrative nature (*e.g.* "When is the second midterm?"), post them here. The sign up link is https://piazza.com/ucla/winter2020/math118/home. We (*i.e.* your instructor and T.A.) will be checking Piazza regularly, and will try our absolute best to ensure that all questions are answered within 24 hours. If you have questions about your grades, or other questions/comments of a personal nature, you should

Tentative Schedule		
Week	**Date**	**Sections**
1	Mon 01/06	Matrix algebra and norms (E. 2)
	Wed 01/08	Linear Systems and Least Squares (E. 3)
	Fri 01/10	Orthogonal matrices, QR factorization, application to least squares (E. 4 & 5)
2	Mon 01/13	Singular Value Decomposition and PCA (E. 6)
	Wed 01/15	Reduced Rank Least Squares (E. 7)
	Fri 01/17	Tensors and HOSVD (E. 8)
3	**Mon 01/20**	**Martin Luther King Jr. day—no class**
	Wed 01/22	k-Means and non-neg. matrix factorization (E. 9)
	Fri 01/24	Graphs and spectral clustering (E. 10 & 12)
4	Mon 01/27	Classifying handwritten digits using SVD and tangent distance (E. 11)
	Wed 01/29	Text preprocessing and mining (E.12)
	Fri 01/31	Introduction to linear programming (C.& Z. 15)
5	Mon 02/03	The simplex method for linear programs (C. & Z. 16)
	Wed 02/05	Dual linear programs (C.& Z. 17)
	Fri 02/07	Catch-up/ review
6	**Mon 02/10**	**Midterm**
	Wed 02/12	Non-simplex methods (C. & Z. 18)
	Fri 02/14	Introduction to convex optimization (C. & Z. 22)
7	**Mon 02/17**	**President's Day—no class**
	Wed 02/19	Constrained optimization: KKT condition (C. & Z. 21)
	Fri 02/21	Constrained optimization: algorithms (C. & Z. 23)
8	Mon 02/24	Introduction to dynamic programming (H. & L. 11.1)
	Wed 02/26	Characterizing problems solvable by dynamic programming (H. & L. 11.2)
	Fri 02/28	Deterministic dynamic programming (H. & L. 11.3)
9	Mon 03/02	Pagerank and Markov Chains (E. 12)
	Wed 03/04	Introduction to integer programming (H. & L. 12.1–12.5)
	Fri 03/06	Branch-and-bound for integer programming (H. & L. 12.6)
10	Mon 03/09	Introduction to Neural Networks (C. & Z. 13.1)
	Wed 03/11	Logistic Regression
	Fri 03/13	The backpropagation algorithm (C. & Z. 13.3)
11	**Fri 03/20**	**Final Exam**

contact me or your T.A. via email. Office hours or immediately after class are also good times to ask questions.

Grading: Your grade will be determined using the following formula: 30% Homework + 30% Midterm + 40% Final exam. You may earn up to 3% extra credit by asking and answering questions on Piazza (see: "Communication").

Homework: Homework will be assigned roughly every week, via Piazza. You are to submit your homework using GradeScope. We strongly encourage you to use word processing software such as LaTeX to type your solutions. **Late homework will not be accepted**, but your lowest homework score will be dropped. Homework problems will be a mix of pen-and-paper calculations and programming exercises. You are encouraged to discuss homework questions on Piazza or in person. However copying other's solutions or programs is considered a serious violation, and you should never share your written solutions with anyone.

Software: For programming exercises you may use either MATLAB or Python.

Exams: There will be one in-class midterm exam and a cumulative final. The midterm will be conducted during the regularly scheduled class period. Make-ups for the final and midterm are permitted only under exceptional circumstances, as outlined in the UCLA student handbook. **Note that you must take the final exam in order to pass this class.** The exams are scheduled for the following dates:

- Midterm: Monday, February 10th
- Final Exam: Friday, March 20th (week 10)

Note that both exams will be of the pen-and-paper variety, but you may be asked to write pseudocode or implement, by hand, a few steps of an algorithm on a toy example.

Classroom Expectations

- Attendance, for class and discussion sections, is mandatory.
- On exams I expect you to give legible, well-justified solutions. A numerical answer without any supporting calculations is unlikely to receive credit, nor is a solution that I can't read.
- As computation is a large part of this course, you are allowed to have your laptop out during class. However, I ask that as a courtesy to your instructor and classmates you keep your phone in your bag, on silent, for the duration of class.
- In this classroom you can expect to be treated with respect, regardless of your age, background, beliefs, ethnicity, gender, gender identity, gender expression, national origin, religious affiliation, sexual orientation and other visible and non-visible differences. All members of this class are expected to contribute to a respectful, welcoming and inclusive environment for every other member of the class.

How to succeed in this course

- Preparing for class by reading ahead in the book (see the course outline) can really help you to get the most out of lectures. Spending even 15 minutes skimming a section before it is taught in class can be useful!
- Office hours are time that we (your instructor and T.A.'s) set aside in order to discuss math with you. You can drop in at any time during office hours. They are a great time to discuss: homework problems you are stuck on, things you didn't quite understand in class, strategies for preparing for or taking tests or going over tests after they have been returned. We may also be able to help you debug your code
- Forming a study group can be very helpful. Remember that often the best way to fully understand a concept is to try to explain it to someone else.
- Make use of Piazza! (See "Communication" section). Again, answering other students questions is a great way to make sure you really understand a concept.

Academic Integrity: From the office of the Dean of Students:

> *With its status as a world-class research institution, it is critical that the University uphold the highest standards of integrity both inside and outside the classroom. As a student and member of the UCLA community, you are expected to demonstrate integrity in all of your academic endeavors. Accordingly, when accusations of academic dishonesty occur, The Office of the Dean of Students is charged with investigating and adjudicating suspected violations. Academic dishonesty, includes, but is not limited to, cheating, fabrication, plagiarism, multiple submissions or facilitating academic misconduct.*

Students are expected to be aware of the University policy on academic integrity in the UCLA Student Conduct Code*. In particular, please note the sections on (1) cheating, (2) plagiarism, and (3) unauthorized study aids.

Accessibility: Students needing academic accommodations based on a disability should contact the Center for Accessible Education (CAE) at (310)825-1501 or in person at Murphy Hall A255. In order to ensure accommodations, students need to contact the CAE within the first two weeks of term.

Notice about sexual harassment, discrimination and assault: Title IX prohibits gender discrimination, including sexual harassment, domestic and dating violence, sexual assault and stalking. Students who have experienced sexual harassment or sexual violence can receive **confidential** support and advocacy from a CARE advocate:

> The CARE Advocacy Office for Sexual and Gender-Based Violence
> 1st Floor, Wooden Center West
> `CAREadvocate@caps.ucla.edu`
> (310) 206-2465

You can also report sexual violence or sexual harassment directly to the University's Title IX Coordinator:

*`http://www.deanofstudents.ucla.edu/Portals/16/Documents/UCLACodeOfConduct_Rev030416.pdf`

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